



MOKO SMART

IN100 Solution Beacon

User Guide

V1.4_2025.07.04

Revision History

Version	Data	Notes	Contributor(s)
V1.0	27-June-2023	Initial version	Leo Wang
V1.1	19-July-2023	Revise the description regarding the default broadcast format.	Leo Wang
V1.2	09-Jan-2024	Add N1 Stamp product model and related description.	Leo Wang
V1.3	08-July-2024	1. Add the upgraded version of UT2 product model and relevant description. 2. Revise the default broadcast format explanation, 3. Add a section on power on/off instructions.	Leo Wang
V1.4	04-July-2025	1. Delete delevant description of UT2 2. Add N2 version description	Leo Wang

Description of Professional Terms

Table 1: Description of Professional Terms

Abbreviations	Full Name	Instructions
Tx Power	Transmission power	-
ADV	advertising	
W	Write	Writable property
R	Read	Readable property
N	Notify	Notify property
RFU	Reserved for use	

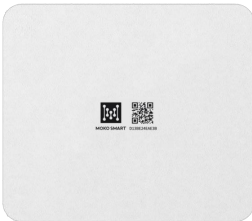
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About This User Guide

This user guide provides instructions for the use of IN100 chip solution devices and supports the configuration of broadcast modes and parameters. It will help sales operations or customers understand how equipment support works and provide a reference for their work.

This document applies to the following products:

Product model name	ID picture
M1I	
M4I	
UT1/UT1P	
N2	
N1 Stamp	

1. About IN100 Chip Solution Features

The main features of products based on IN100 chip solutions include the following aspects:

- Bluetooth Low Energy 5.3 Compliant
- **Support iBeacon, Eddystone(UID/TLM), Customized frame**
- **Supports a maximum of two broadcast channels**
- **Support Continuous advertising mode or Event-trigger advertising mode (button trigger)**
 - **UT1 / UT1P** - support continuous advertising mode and button trigger mode
 - **M1I** - support continuous advertising mode;
 - **M4I** - support continuous advertising mode;
 - **N1 Stamp** - support continuous advertising mode
 - **N2** - support continuous advertising mode
- **The broadcast parameters of the device can be configured only before the device leave the factory, and cannot be configured again after the device leave the factory**
- **Unconnectable** (The device does not support the connection of other Bluetooth devices)
- Low cost and high performance
- Enhanced privacy mode support
- 4KB eFuse memory & 4KB SRAM
- Programmable TX output power. **N2&N1 Stamp&M1I&M4I support up to +4dBm; UT1 support up to 0dBm.**
- Integrated low leakage LDO, Single cell 1.5V/3V battery support

Sub-uW power consumption for multi-year operation on a tiny battery, Sleep mode < 650nA with 32kHz RC ON

2. Bluetooth Broadcast Data Format

2.1 Default Setting

The IN100 Series tags are used in the same configuration, the broadcast parameters are as follows:

SLOT1
<i>Adv type -- NanoBeacon (MOKO Custom)</i> <i>Adv interval -- 3000ms</i> <i>Adv Tx Power -- 0dBm</i> <i>Adv mode -- Continuous</i>

You can refer to the tables below for the actual data format of the device.

Advertisement Data (NanoBeacon)			
Byte offset	Field	Example value	Descriptions
0	Data length	0x06	AD length of Flags content
1	Data type	0x09	AD type: Flags
2 - 6	Device name	0x4e 42 54 41 47	Device name : NBTAG
7	Data length	0x12	AD length of NanoBeacon content
8	Data type	0xff	Manufacturer Spec
9 - 10	Manufacturer ID	0x05 05	Manufacturer ID
11	Adv type	0x00	0x00 -- Normal Adv type
12	Battery voltage	0x60	(Little-endian), uint: 0.03125V Content Parse: 0x60 >>>96 -- > 96×0.03125V = 3V
13 - 14	Beacon temperature	0x30 0A	(Little-endian), uint: 0.01℃ Content Parse: 0x0A 30>>>2610 -- > 2610×0.01℃ = 26.1℃
15 - 18	SEC_CNT	0x8B 7C 3B 00	(Little-endian), SEC_CNT is a 1 second resolution counter that represents time since beacon power-up or reboot. If this value is reset (e.g. on a reboot), the ADV count field must also be reset. 4 bytes, Content parse: 0x00 3B 7C 8B>>>3,898,507, that is 3,898,507s
19	Trigger status	0xDD	Fixed value: 0xDD
20-25	MAC address	0x46 4D A3 71 3E FF	(Little-endian), Content parse: Beacon MAC address: FF:3E:71:A3:4D:46

3. How to power ON/OFF the tag

3.1 How to power ON/OFF the tags?

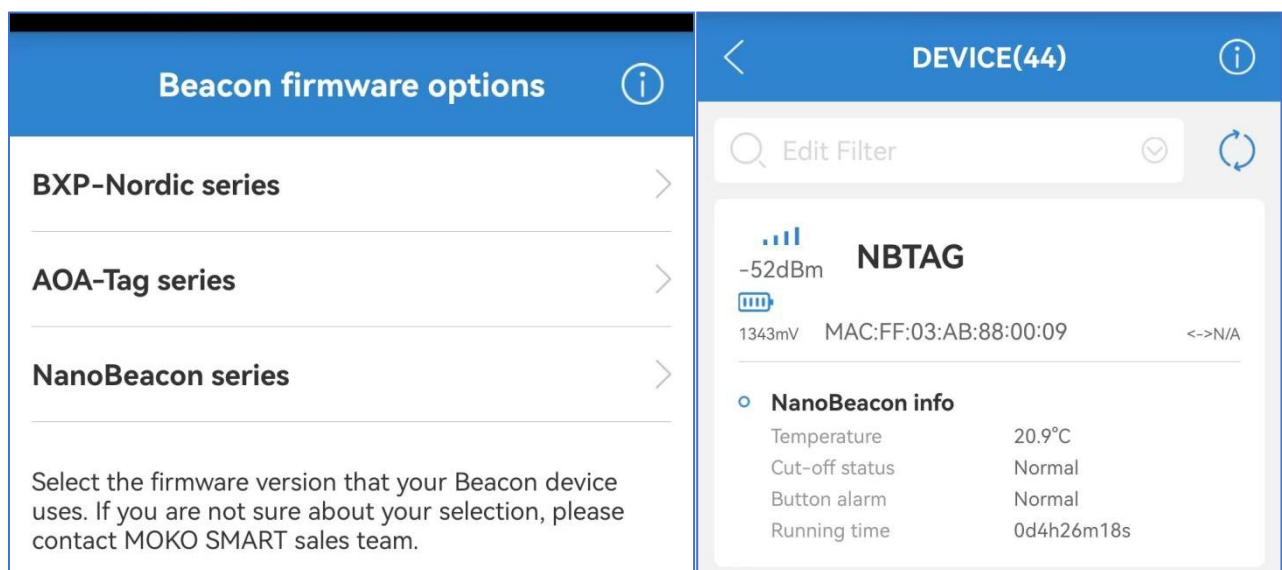
Please check the table below for a description of the features

Product model	Power on	Power off
M1I	The device is powered on by default. Open the device casing and remove the insulating sheet to power it on.	Power-off function can be achieved by installing an insulating sheet.
M4I	The device is powered on by default. Remove the insulating sheet to power it on.	Power-off function can be achieved by installing an insulating sheet.
UT1/UT1P	The product is powered off by default. Tear along the easy tear line to power on the product.	Once powered on, it does not support powering off again.
N1 Stamp	The device is powered on by default. Remove the insulating sheet to power it on.	Power-off function can be achieved by installing an insulating sheet.
N2	Peel off the release paper on the back of the product to power on the device.	Not support

3.2 How to confirm the tag is powered on?

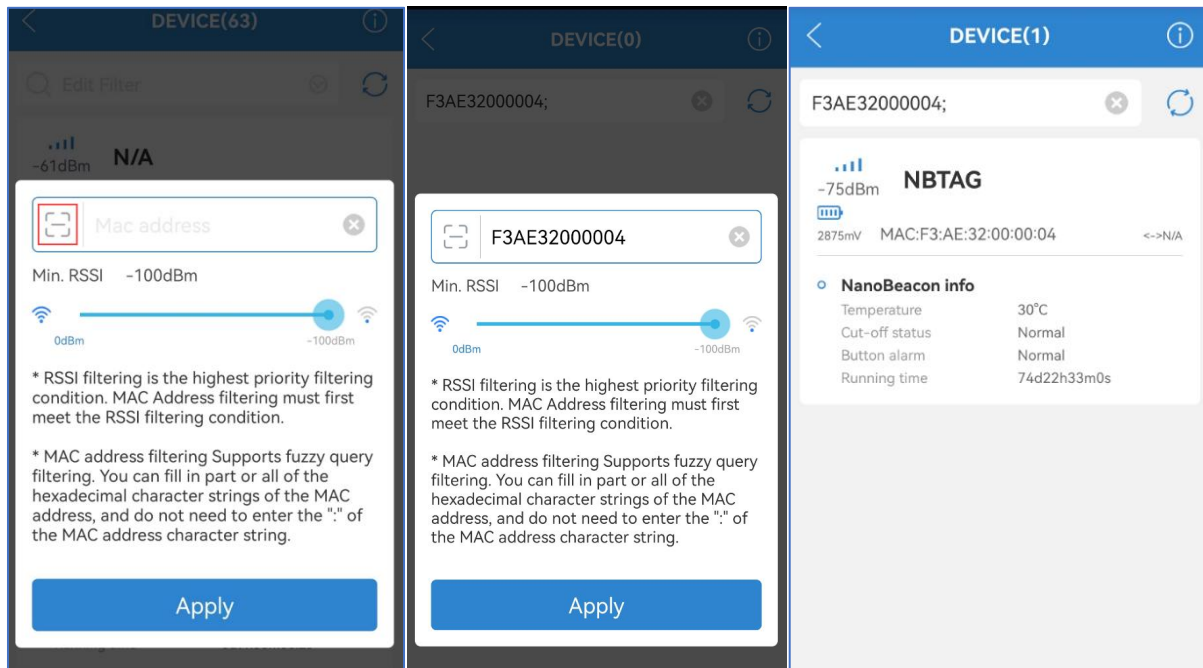
You can determine if the device is operational by using the BeaconX Pro app to scan for its broadcast. If the broadcast from the tag's specified MAC address is detected, it indicates the device is powered on. If no broadcast is detected after an extended period of scanning, it can be inferred that the device is not yet powered on.

Step1. Open the BeaconX Pro app, with app version V2.0.7 or higher, and select the “*NanoBeacon series*”, then you will enter the scanning page.



Step2. Click on “*Edit Filter*” to enter the scan filter page. Choose the QR code scanning function and use your phone camera to scan the QR code on the product. The QR code contains the MAC address information. Once the QR code

is successfully recognized, the MAC address information of this device will automatically populate the scan filter box. Click '**Apply**' and wait for a moment. If the app's scan page detects the broadcast information from the corresponding device, it indicates that the device has been successfully powered on and is operating normally.

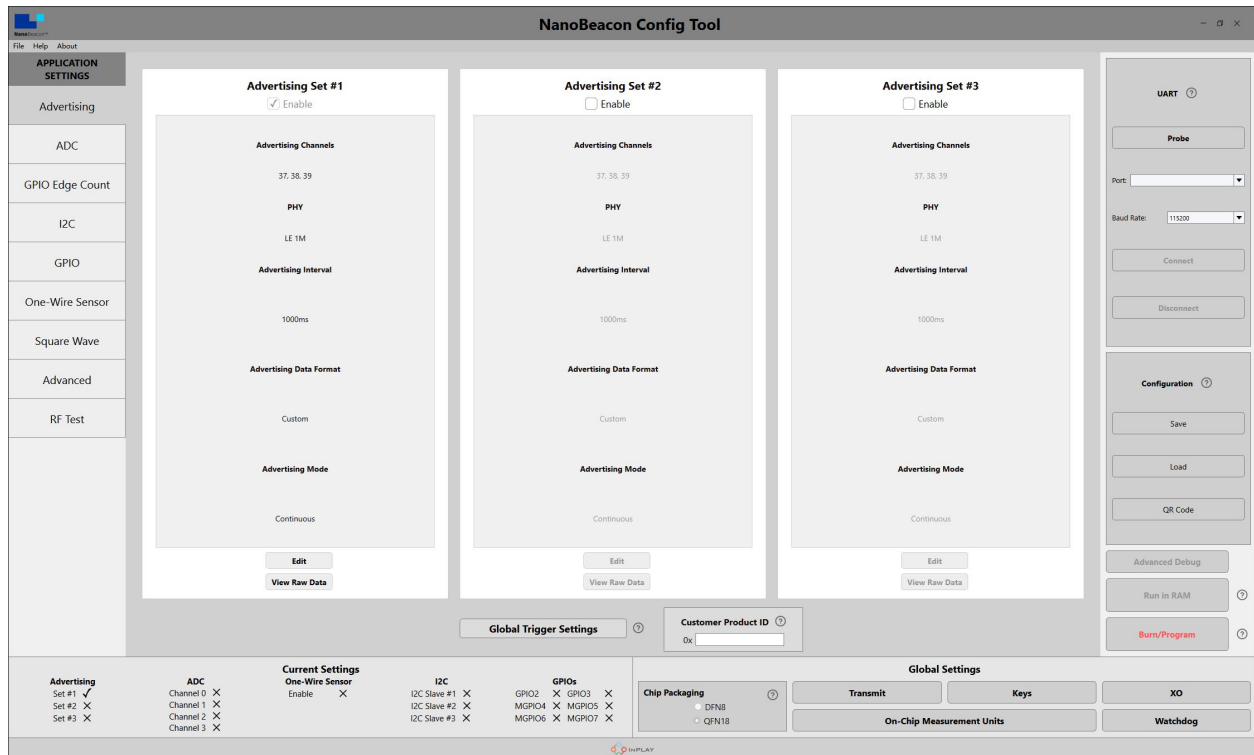


4. Advertising Data Configuration

The hardware based on the IN100 chip solution support configure the advertising data and advertising parameters of the Bluetooth beacon according to customer's application requirements through the configuration tool to achieve the purpose of different broadcast functions.

MOKO can also provide pure hardware equipment, which can be configured by the customer, or MOKO can configure the parameters as customer's requirements for verification.

The GUI of the configuration tool is shown in the below picture.



Note: The broadcast parameters of the device can be configured only once before the device leave the factory, and cannot be configured again after the device leave the factory

The Bluetooth beacon supports configure the following functions and parameters:

- The Bluetooth beacon supports up to two advertising slots sets which means beacon can broadcasts different advertising packets separately.
- The Bluetooth beacon supports two commonly used advertising data formats.
 - iBeacon: Apple proprietary Beacon advertising frame format based on Bluetooth standard advertising (The parameters of UUID/ Major/ Minor/ Measured Tx Power in the iBeacon data format support configure)
 - Eddystone(UUID/TLM): Google proprietary Beacon advertising frame format based on the service data of the Bluetooth standard advertising.
 - Custom: the advertising data in a custom user-defined format as customer's application requirements
- The Bluetooth beacon supports configure the button trigger function which means the device support advertising the specified slot's broadcast after trigger event occurs.

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